Context-Aware Music Companion Using Wearable Health Data

Choosing the right song, from your own Spotify library, for the moment you are in

Clément Royer
Dept. Computer Science
Hanyang University
Seoul, S. Korea
clement.royer@edu.devinci.fr

Cyprien Vanden Abeele
Dept. Computer Science
Hanyang University
Seoul, S. Korea
cyprien.vanden_abeele@edu.devinci.fr

Lucas Miskulin
Dept. Computer Science
Hanyang University
Seoul, S. Korea
lucas.miskulin@edu.ece.fr

Riad Mekideche
Dept. Computer Science
Hanyang University
Seoul, S. Korea
riad.mekideche@edu.devinci.fr

***Abstract*—Waking up, stress, an afternoon energy dip and falling asleep are moments in which the right music can help, but nobody wants to search for it at that moment. We propose an iPhone app that picks songs from the user's own Spotify library to fit the moment. The app gives each saved song two simple scores (how calm or energetic it is, and how positive or negative it sounds). It also estimates, from smartwatch data (heart rate, sleep, steps), the time of day and the weather, whether the person is more or less activated than usual. It then plays a short queue that starts close to the person's current state and gradually moves toward a calmer or a more energetic state. Instead of asking each user to train the app with weeks of personal data, we use existing public datasets, and we state clearly how accurate such models are. This document presents the user needs, the related software, our design choices and the requirements.**

Keywords—music recommendation, wearable data, context-aware app, Spotify, iOS

I. Role Assignments

Table I gives the roles. The project has two main programmers, so the Software developer role is shared: one member builds the app, the other builds the data and AI part. Roles are set for the first half of the term and can be rebalanced afterwards; every member also reviews the others' work.

TABLE I. Role Assignments

Role	Name	Task Description
User	Cyprien Vanden Abeele	Uses the app every day with a Spotify account and a smartwatch; notes whether the songs fit the moment and why a song was skipped; reports bugs; runs the small pilot test.
Customer	Cyprien Vanden Abeele	Checks that each feature answers the needs in Section II-B; approves mock-ups and demos; decides trade-offs such as privacy versus accuracy.
Software developer	Riad Mekideche	Builds the iPhone app: Spotify sign-in and library import, reading of watch data, screens and playback.
Software developer	Lucas Miskulin	Builds the data and AI part: song scores, state estimate, song-choice rules and the evaluation scripts.
Development manager	Clément Royer	Plans the milestones, assigns tasks, manages the GitHub repository and code reviews, writes the documents, and handles the Apple and Spotify developer accounts.

II. Introduction

A. Motivation

The need. Poor sleep and stress are an everyday problem in Korea. A report on the Korean sleep market, based on a KB Financial Group think-tank study, gives an average sleep time of 471 minutes per day, 30 minutes below the OECD average. It also cites a Philips survey of 13,000 people in which only 41% of Koreans were satisfied with their sleep (55% on average over 13 countries) and stress was the first cause of disturbed sleep (54%) [1]. The survey date is not given in the article.

Why music. A Cochrane review of 13 randomised trials with 1,007 adults compared listening to music with no treatment or usual care. In ten of these trials (708 participants), music improved sleep quality by 2.79 points on the Pittsburgh Sleep Quality Index (lower is better; 95% CI −3.86 to −1.72), with moderate certainty of evidence. The participants knew they were receiving music, which can bias the result [2]. This is evidence about music in general: it does not show that our app will improve well-being, and our pilot will not test that.

Why now. Music streaming is part of daily life, yet the personalisation offered by streaming platforms still ignores the body of the listener (Section II-C).

Target users. Our first users are students and young workers who listen to music on their phone every day, wear a smartwatch and have irregular schedules: early classes, long afternoons and late study nights. They do not want a new habit or a long setup; they want the music they already like, at the right moment, with one tap. The team members belong to this group, which is why they can act as users during development (Table I).

B. Problem Statement (User Needs)

Given the time of day, the recent signals of the person's body compared with their own usual values, and the situation they want help with, *which songs from their own library should be played, and in which order, to help them through this moment?* The target users already stream music every day

and face recurring low-energy or high-stress moments. Table II lists the four situations we address.

TABLE II. TARGET SITUATIONS

Situation	Typical moment	Goal of the music
Wake-up	Morning, after sleep	Gently more energetic
Stress	Any time of day	Calmer
Low energy	Afternoon slump	More energetic
Wind-down	Evening, night	Calm and sleepy

The needs behind this problem are the following.

- **N1. No effort.** A tired or stressed person does not want to search for the right playlist. One tap should start suitable music.
- **N2. My own music.** Songs must come mainly from the user's own library, and never from generated or generic audio.
- **N3. Fit to my body and my day.** Suggestions based only on listening history and time of day do not use how the person is doing physically.
- **N4. Control of my data.** Heart rate and sleep are sensitive. The user must know what is read and be able to delete it.

C. Existing Solutions

Table III compares the closest products we found. The facts come from the cited pages; we did not install or test the products.

TABLE III. COMPARISON WITH RELATED SOFTWARE

Solution	Uses body signals?	Plays the user's own songs?	Notes
Endel [3]	Yes: heart rate, time of day, weather, location	No: it generates its own sounds	Paid subscription; iOS, Android, Apple Watch
Spotify daylist [4]	No: listening habits and time of day only	Through the listening history	Playlist updated several times a day
Spotify Prompted Playlists [5]	No: typed prompt and listening history	History used by default	Premium, US and Canada, beta (Jan. 2026)
Proposed app	**Yes: heart rate, sleep, steps, time, weather**	**Yes: the saved Spotify library first**	**iPhone and Spotify first; other services later**

Apps such as Endel react to the body but create their own audio, and the vendor's claims about its effects are company claims [3]. Streaming platforms personalise from history and context, but the articles describing daylist and Prompted Playlists mention no wearable or sensor [4], [5]. We found no product that combines body signals, daily context and the user's own library to choose music; this rests on a limited search, and Spotify could add such a feature. Our argument is therefore not better accuracy than Spotify, but the added value of body signals and of a clear goal: helping through a situation.

Research. A systematic review of context-aware music recommenders (783 records screened, 100 papers included) finds that time and location are the most used signals, while body signals such as heart rate are less studied, partly because suitable datasets are lacking [6].

D. Proposed Solution

Overview. Fig. 1 shows the app in three parts. (1) *Song scores:* each song of the library gets an energy score (calm to energetic) and a positivity score (negative to positive). (2) *State estimate:* from the watch, the time and the weather, a small model says whether the person is currently more or less activated than usual. (3) Song choice: simple rules pick a short queue from the library, completed by MuSe songs when needed. The person can always choose the situation with one tap and skip any song. AI is used in the first two parts: a language model for song scores and a trained model for the state estimate.

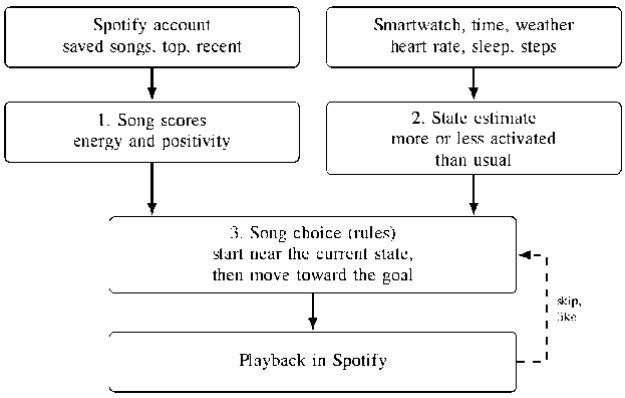


Fig. 1. System overview. The dashed arrow shows the feedback from the listener.

An example. It is 11:30 p.m. and the user taps Wind-down. The watch shows a short previous night and a heart rate above the person's usual evening value, so the state estimate is "more activated than usual". The app therefore does not start with the calmest song of the library: the first two songs have a medium energy, and the next ones become calmer step by step. If the user skips a song, the next one is taken from the same energy level, and the skip is recorded for future queues.

What "training" means here. A model learns patterns from examples. Asking every user for weeks of personal data before the app works would make a poor first day. We therefore train on public datasets in which other people wore a watch and rated their mood. The app then compares each user with their own average, which needs no training, only a short period of watch data (we plan 7 to 14 days).

Part 1: song scores. Spotify removed its audio features (including valence and energy) and 30-second previews for new apps in November 2024 [7], so the app cannot analyse the sound itself. We use MuSe, a public dataset of 90,408 songs with valence, arousal (energy) and dominance scores built from Last.fm user tags; Spotify IDs are given for about 68% of its songs and are described as tentative [8]. MuSe is small compared with Spotify's catalogue, so only part of a library will match. For the other songs, a large language model (LLM) estimates the two scores from the title and the artist. We will check this estimate on songs present in MuSe, and measure the share of the team's own libraries that MuSe covers during the first weeks.

Part 2: state estimate. The inputs are the difference between recent heart rate and heart-rate variability (small changes in the time between heartbeats) and the person's own average, last night's sleep, steps and the time of day. The weather is only a secondary input. The output is "more or less activated than usual", never a named emotion. The model is

trained on two open datasets: 21 students wearing a Samsung watch and an Oura ring who reported their feelings [9], and a Fitbit dataset with self-reported mood ratings [10]. Watch brands measure some values differently, so we only use values that every brand provides, expressed as a difference from the person's own average. Training on one dataset and testing on the other is our way to check that the model works across brands.

How the estimate is computed. For each signal, the app computes a z-score: the recent value minus the person's usual value, divided by its usual variation. A positive heart-rate score and a negative heart-rate-variability score both point to a more activated state. These scores, last night's sleep compared with its usual value, the steps of the last hour and the time of day form the model input. We will start with a small, explainable model (logistic regression, then gradient-boosted trees if they do better on held-out people) that returns a value between 0 (much calmer than usual) and 1 (much more activated than usual). Until the personal average is ready, the app uses only the time of day and the chosen situation.

Apple Watch and expected accuracy. We found no public dataset that links Apple Watch signals to real-life mood, so a short pilot on Apple Watch will measure the remaining gap.

Real-life studies report modest results, for example 60.4% balanced accuracy for a yes/no anxiety state from heart rate alone [11]. Our model will not be perfect either. It only sets the start of the queue, and the user has the final word.

Part 3: song choice. The rules first keep songs the person likes (saved songs, top and recently played tracks). Each situation has a goal: more or less energy. The queue has 3 to 5 songs: the first one or two have an energy close to the person's current state, and the next ones move step by step toward the goal. This follows the "iso principle" of music therapy: in a controlled experiment, adults made sad by a film clip reported more positive affect after a sad-then-happy pair of pieces than after other orders, although not all group differences were significant [12]. We limit the matching phase to two songs, so that music matching a low mood does not keep the person in that mood. Skips and likes are used to avoid songs the person dislikes.

How the queue is built. Energy and positivity scores are put on the same scale from 0 to 1. The energy of the first song is the state estimate, the energy of the last song is the goal of the situation (for example 0.2 for wind-down and 0.7 for wake-up), and the songs in between are spaced evenly. For each position, the app picks the library song closest to the target energy, prefers positive songs for the stress and low-energy situations, and avoids repeating an artist or a song played in the last days. These rules are simple on purpose: each choice can be explained to the user (FR11) and tested on recorded data (FR12).

When the library is not enough. A library may not hold enough suitable songs: MuSe covers only part of it, and a library built for workouts may contain almost no calm songs for the wind-down situation. In that case the app completes the queue with MuSe songs that have a Spotify ID. These songs already have energy and positivity scores, so no estimate is needed. To stay close to the person's taste (N2), the app keeps MuSe songs whose genre or artists also appear in the library, and checks that each Spotify ID points to the right title and artist before playing it, since MuSe describes these IDs as tentative [8]. Library songs always come first, at most two songs of a queue are suggestions, and they are marked as new. A like saves the song to the library; a skip removes it from future suggestions. The pilot will measure how often this fallback is needed and compare the skip rate of these songs with that of library songs.

What the user sees. At first launch, the user signs in with Spotify and chooses which Apple Health data the app may read, with one sentence explaining each choice. The home screen shows the four situations as large buttons, with the suggested one highlighted. After one tap, the queue starts and the screen shows the cover art of the current song, the reason for the queue and three actions: skip, like and "did this fit?". A settings screen lists the stored data and lets the user delete it (FR13).

Data and privacy. Health data are read from Apple Health and processed on the phone; the app stores daily summaries and the model output, not raw heart-rate series. The library, the song scores and the feedback are also stored on the phone. The only data sent outside are the Spotify requests and, for songs missing from MuSe, the title and artists sent to the LLM (NFR1). Disconnecting Spotify or deleting the data removes everything the app has stored.

Technical choices. The app is written in Swift with SwiftUI. Apple Health data are read with HealthKit; Spotify is accessed through its Web API for the library and through the Spotify iOS SDK for playback. The state model is trained in Python with scikit-learn on the public datasets, then converted to Core ML so that it runs on the phone and health data never leave it. MuSe is stored in the app as a small table (title, artists, scores, Spotify ID), so the song scores are available offline. The LLM estimate is requested once per song and cached, which keeps its cost and delay low.

Options we considered. Table IV shows the main alternatives and why we chose our approach.

TABLE IV. OPTIONS CONSIDERED AND OUR CHOICE

Question	Options	Our choice and why
Is a song calm or energetic?	Spotify audio features (removed for new apps [7]); analyse the audio ourselves (Spotify does not provide it); learn from each user's listening history (weeks of data needed).	**MuSe scores, then an LLM estimate.** Public, free, works on the first day.
What is the person's state?	A camera reading the face (needs a photo each time; not usable in bed or on the move); only ask the user (simple but not smart); a model trained on each user's own data (poor first days).	**Model trained on public watch datasets, compared with the person's own average, plus a one-tap choice of the situation.**
Which songs to play?	Generate new music (not the user's songs, see N2); Spotify recommendations (removed for new apps [7]); the user's own library with simple rules.	**Own library with rules, completed by MuSe songs when needed.** Matches need N2 and is easy to explain and test.

How we will test it. (1) Song scores: compare the LLM estimate with MuSe on songs present in both, and report the share of the library that MuSe covers. (2) State estimate: train on one dataset and test on the other, and test on people the model has not seen. Random splits of the same person's data give results that are too optimistic. We compare the model with simple baselines such as the time of day alone. (3) Pilot:

at most five Spotify accounts for two to three weeks (the limit of Spotify's development mode [13]), measuring the skip rate and a one-tap "did this fit?" answer. On some days chosen at random, the app plays a queue built from the time of day only, so that the skip rates and answers of the two kinds of queue can be compared. The pilot measures fit, not health benefits.

What counts as success. For the song scores, we report the correlation between the LLM estimate and MuSe and the mean error on each scale. For the state estimate, we report balanced accuracy on held-out people and compare it with the time-of-day baseline; if the model does not beat the baseline, it is not used and the app relies on the situation choice alone. For the pilot, the main measures are the share of songs skipped in the first 30 seconds and the share of positive "did this fit?" answers, compared between the two kinds of queue.

Limits and risks.

- Spotify's development mode allows five users, and the owner needs Premium [13]. The app is a demonstrator, not a public release.
- Spotify changed its Web API again in February 2026; we will first confirm with a live test account that saved songs, top items and recent plays are still available. If an endpoint is no longer available, the app falls back on saved songs only, or on the demo mode (FR12).
- Playback needs Spotify Premium and a physical iPhone with the Spotify app installed. If playback fails, the songs open in the Spotify app (FR9).
- Apple Health data cannot be read while the iPhone is locked, so the app reads it when opened.
- The datasets are small and the expected accuracy is modest (see above). The one-tap situation choice keeps the app useful even when the estimate is wrong.
- The LLM may give wrong scores for unknown or ambiguous songs. Its estimates are checked on MuSe songs, and a song that is often skipped in a given situation is moved down in future queues.
- The app is a wellness companion: it does not diagnose anything.

Why Spotify and Apple Watch, and how to go further. We start with Spotify and the Apple Watch because they are what the team owns and uses every day, which makes development and the pilot possible from the first week. This is a practical choice, not a limit of the design. All music access goes through one small interface (sign in, import the library, play a queue) and all health data through another (heart rate, heart-rate variability, sleep, steps). The song scores, the state estimate and the queue rules never call Spotify or Apple Health directly. Supporting Apple Music or Deezer therefore means writing one new music connector, and supporting a Galaxy Watch or a Fitbit on Android means writing one new health connector, for example through Health Connect. The model does not need to change: it is trained on Samsung, Oura and Fitbit data and only uses differences from the person's own average, which every brand can provide. The only extra step for another music service is to match MuSe songs by title and artist, since the Spotify IDs of MuSe are specific to Spotify.

E. Development Plan

The project runs until December 2026. Each milestone ends with a demo to the customer (Table I), who decides what to change before the next one.

- **M1, end of October:** Spotify sign-in, library import, reading of Apple Health data, demo mode with recorded data.
- **M2, mid-November:** song scores from MuSe and the LLM; first state model trained on one dataset and tested on the other.
- **M3, end of November:** queue building with the MuSe fallback, playback, feedback, explanation and data-control screens.
- **M4, December:** pilot with up to five accounts, evaluation against the time-of-day queues, final report and demo.

Way of working. The team meets once a week to review progress against the milestones. All code goes through pull requests on GitHub and is reviewed by another member before it is merged. The data and AI part is first developed on recorded data, so that it does not wait for the app, and the two parts are joined through the interfaces described in Section II-D. The riskiest points come first: Spotify access is checked with a live test account in M1, and the cross-dataset test of the state model is done in M2.

III. Requirements

Priorities: **Must**, **Should**, **Could**.

A. Functional Requirements

The app shall:

- **FR1. Spotify sign-in (Must).** Let the user connect a Spotify account through Spotify's secure sign-in (OAuth), and disconnect at any time.
- **FR2. Library import (Must).** Download all saved songs (title, artists, cover art), page by page, and store them on the phone.
- **FR3. Watch data (Must).** With the user's permission, read heart rate, heart-rate variability, sleep and steps from Apple Health when the app is opened.
- **FR4. Personal average (Must).** Compute the person's own usual values over 7 to 14 days and compare recent values with them.
- **FR5. Situation choice (Must).** Offer the four situations of Table II as one-tap buttons, and suggest one from the time and the state estimate (Should).
- **FR6. Song scores (Must).** Give each song an energy and a positivity score from MuSe, and from an LLM estimate when the song is not found (Should).
- **FR7. State estimate (Must).** Estimate whether the person is more or less activated than usual with the trained model; the result is never shown as an emotion label.
- **FR8. Queue building (Must).** Build a queue of 3 to 5 songs from the library that starts near the current state and moves toward the goal, with at most two matching songs. If the library has too few suitable songs,

complete the queue with at most two MuSe songs available on Spotify (Should).

- **FR9. Playback (Must).** Start the queue in Spotify; if playback is not possible, open the songs in the Spotify app.
- **FR10. Feedback (Must).** Record skips, likes and a "did this fit?" answer, and avoid songs the person dislikes.
- **FR11. Explanation (Should).** Show a short reason for the queue, for example "short night, morning: calm songs first, then more energy".
- **FR12. Demo mode (Should).** Run the app on recorded data, so demonstrations and tests do not need a live watch.
- **FR13. Data control (Must).** Let the user view, export and delete all stored data.
- **FR14. Other services (Could).** Support another music service or another watch through the same interfaces.

B. Non-Functional Requirements

- **NFR1. Privacy.** Health data stay on the phone. The LLM estimate sends only song titles and artists, never health data. The user gives explicit consent for each data source.
- **NFR2. Security.** Spotify tokens are kept in the iOS Keychain and all network traffic uses HTTPS.
- **NFR3. Speed.** A queue appears within 2 seconds of choosing a situation, and importing 2,000 songs takes under 60 seconds on Wi-Fi (targets to be measured).
- **NFR4. Missing data.** If watch data are missing or old, the app still works using the time of day and the library.
- **NFR5. Ease of use.** From launch to music takes at most two taps.
- **NFR6. Spotify rules.** Show cover art unaltered, display the Spotify logo and a link to Spotify, as required by Spotify's design guidelines.
- **NFR7. Responsible use.** The app is a wellness companion, makes no medical claim, never names an emotion for the user, and always lets the user stop or change the music.
- **NFR8. Testability.** Song scores, state estimate and song choice are covered by automated tests using recorded data (FR12).

REFERENCES

[1] Y. Whan-woo, "Sleep market in Korea reaches W3 tril. says report," *The Korea Times*, Apr. 10, 2024. [Online]. Available: https://www.koreatimes.co.kr/www/biz/2024/04/126_372435.html

[2] K. V. Jespersen, V. Pando-Naude, J. Koenig, P. Jennum, and P. Vuust, "Listening to music for insomnia in adults," *Cochrane Database Syst. Rev.*, no. 8, Art. no. CD010459, 2022, doi: 10.1002/14651858.CD010459.pub3.

[3] Endel, "Endel: personalized soundscapes." [Online]. Available: https://endel.io/ (accessed Oct. 20, 2026).

[4] "Spotify launches ever-changing 'Daylist' playlist that adapts to users' moods," *Music Business Worldwide*, 2023. [Online]. Available: https://musicbusinessworldwide.com/spotify-launches-ever-changing-daylist-playlist-that-adapts-to-users-moods

[5] "Spotify brings AI-powered Prompted Playlists to the U.S. and Canada," *TechCrunch*, Jan. 22, 2026. [Online]. Available: https://techcrunch.com/2026/01/22/spotify-brings-ai-powered-prompted-playlists-to-the-u-s-and-canada/

[6] Á. Lozano Murciego, D. M. Jiménez-Bravo, A. Valera Román, J. F. De Paz Santana, and M. N. Moreno-García, "Context-aware recommender systems in the music domain: a systematic literature review," *Electronics*, vol. 10, no. 13, Art. no. 1555, 2021.

[7] Spotify for Developers, "Introducing some changes to our Web API," Nov. 27, 2024. [Online]. Available: https://developer.spotify.com/blog/2024-11-27-changes-to-the-web-api

[8] C. Akiki and M. Burghardt, "MuSe: The Musical Sentiment Dataset," *J. Open Humanit. Data*, vol. 7, 2021, doi: 10.5334/johd.33.

[9] S. Labbaf *et al.*, "Physiological and emotional assessment of college students using wearable and mobile devices during the 2020 COVID-19 lockdown: an intensive, longitudinal dataset," *Data in Brief*, vol. 54, Art. no. 110228, 2024, doi: 10.1016/j.dib.2024.110228. Data: Dryad, doi: 10.7280/D1WH6T.

[10] "The PMData dataset," Simula Research Laboratory. [Online]. Available: https://datasets.simula.no/pmdata (accessed Oct. 2026).

[11] M. S. Ahmed *et al.*, "WatchAnxiety: a transfer learning approach for state anxiety prediction from smartwatch data," arXiv:2509.13725, 2025.

[12] K. Starcke, J. Mayr, and R. von Georgi, "Emotion modulation through music after sadness induction—the iso principle in a controlled experimental study," *Int. J. Environ. Res. Public Health*, vol. 18, no. 23, Art. no. 12486, 2021.

[13] Spotify for Developers, "Quota modes." [Online]. Available: https://developer.spotify.com/documentation/web-api/concepts/quota-modes (accessed Oct. 20, 2026).